

Operational Diagnostics Report

Nairobi Depot Throughput Investigation | Oil & Gas Distribution Network

Prepared for: Operations Director
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Scope: 4 depots, 100 operating days

1. Context

We reviewed daily fluid distribution throughput across our four regional storage depots — Mombasa, Nairobi, Kisumu, and Eldoret — over 100 consecutive operating days (January–April 2026). The review was triggered by a system-wide drop in average daily throughput and was intended to answer one question for leadership: **is this normal variation, or a problem that needs an operational response?**

2. Insight — Root Cause Identified

The system-wide slowdown is **not a network-wide issue**. It is fully explained by a single depot: **Nairobi**, which experienced a sudden and sustained 40% drop in average daily throughput starting **February 1st, 2026**, and has not recovered since. Mombasa, Kisumu, and Eldoret have operated within their normal historical range throughout the period.

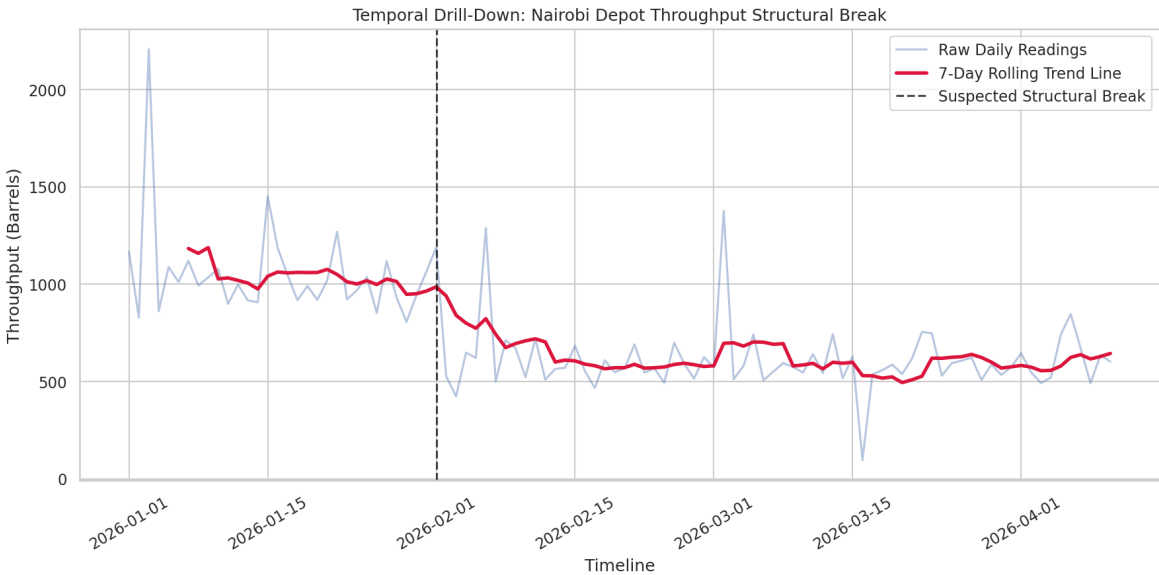


Figure 1: Nairobi's 7-day rolling average throughput falls sharply and permanently at the February 1st mark, indicating a structural break rather than day-to-day noise.

We tested the two most likely alternative explanations — **weather** and **scheduled maintenance** — and ruled both out. Throughput shows essentially zero correlation with ambient temperature ($r = 0.01$) or maintenance activity ($r = -0.02$). This points to a **localized, internal asset failure inside the Nairobi facility** (e.g., a valve blockage or pump degradation) as the most likely cause.

3. Supporting Evidence

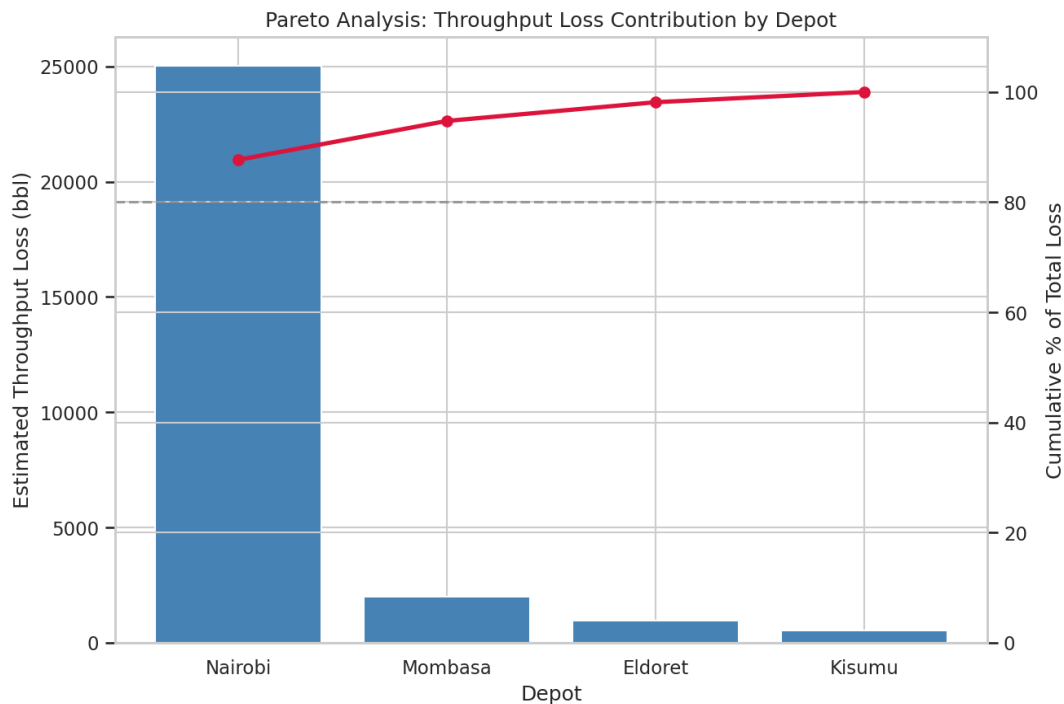


Figure 2: Pareto analysis of estimated throughput loss by depot. Nairobi alone accounts for the overwhelming majority of total network loss — comfortably clearing the classic 80/20 threshold — confirming this is a single-site problem, not a distributed one.

- **Structural break, not noise:** the drop occurs on a single day and persists for the full remaining period — inconsistent with random daily fluctuation.
- **Concentration:** Pareto analysis shows Nairobi is responsible for the large majority of system-wide throughput loss.
- **Weather ruled out:** correlation between throughput and temperature is negligible ($r \approx 0.01$).
- **Maintenance ruled out:** correlation between throughput and scheduled maintenance flags is negligible ($r \approx -0.02$).

4. Recommended Actions

1. **Immediate (this week):** Dispatch an emergency engineering inspection team to the Nairobi depot to physically check for valve blockages, pump wear, or line damage — the most likely internal causes given that external factors have been ruled out.
2. **Short-term (next 2–4 weeks):** Once the fault is confirmed, schedule the necessary repair or part replacement and validate the fix by confirming throughput returns to its ~1,000 bbl/day baseline.
3. **Preventive (ongoing):** Add an automated rolling Z-score alert to the existing ETL pipeline so that any future structural deviation like this one is flagged within 48 hours, rather than discovered weeks later through manual review.

Bottom line: This is a localized, fixable equipment issue at one site — not a systemic network problem. The cost of a short inspection visit is small relative to the throughput already lost, and the longer we wait, the more that loss accumulates.